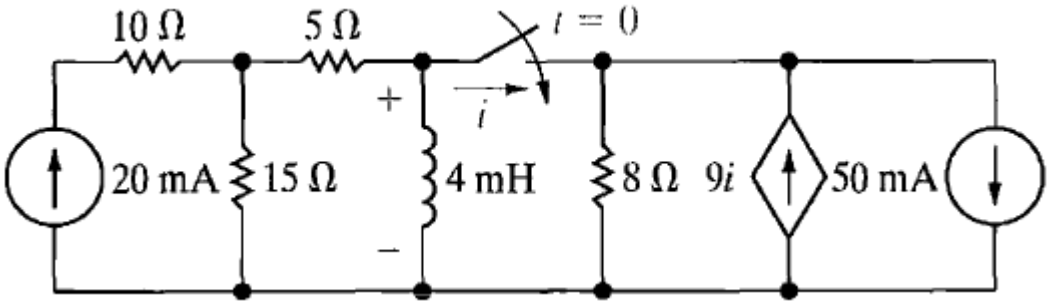
Name:	R. No:	Section:

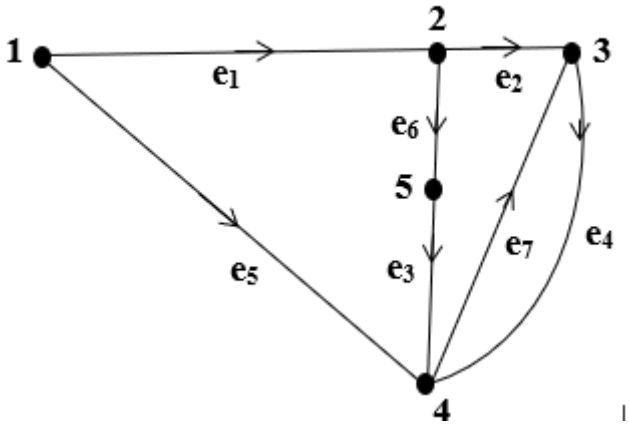
1. The switch in the circuit has been open for a long time before closing at $t = 0$.

(i) Find the current through the inductor $i_L(t)$

(ii) Using the result in (i), find the voltage across the inductor $v_L(t)$ [9+1]



2. Find the fundamental cutset corresponds to the branch e_3 and fundamental circuit matrix corresponds to e_6. Take e_1, e_2, e_3 and e_4 as tree branches. (Give only one row of the Q matrix corresponds to e_3 and row of B matrix corresponds to e_6). [2+1]



e1 e2 e3 e4 e5 e6 e7

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3. Determine values of R1 and R2 in the following circuit so that the Thevenin's equivalent voltage is 1.3 V and Thevenin's equivalent resistance is 0.3 Ω. [1+1]

